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Data Mining Project Report

Introduction

This project implements an interactive supermarket simulation that models and analyzes consumer purchasing behavior using association rule mining. Users can create or import shopping transactions, clean the data, and apply Apriori and Eclat algorithms to discover frequent itemsets and association rules. The system is designed to demonstrate how data mining techniques can uncover hidden patterns to support retail decision-making, such as product placement, cross-selling strategies, and recommendations.

System Overview

The user interface is implemented entirely with Streamlit for easy interaction and visualization. The system allows users to:

- Create transactions manually using a product selector
- Upload a CSV file of supermarket transactions
- Automatically clean and preprocess the dataset
- Run Apriori and Eclat algorithms to mine frequent itemsets
- Compare algorithm performance
- Query rules to view product recommendations
- Generate business insights based on rule confidence

Data Preprocessing

Real-world transaction data is often messy and inconsistent. The preprocessing section removes empty transactions, single-item entries, invalid or unknown products, duplicates, inconsistent casing, and trims whitespace. A cleaning report is generated summarizing the removed and standardized items.

Issues Detected & Cleaned:

Issue Type	Action Taken	Count Removed
Empty transactions	Removed	5
Single-item transactions	Removed	6
Duplicated items	Deduplicated	9
Case inconsistencies	Standardized to lowercase	All
Invalid items (not in product catalog)	Removed	2

Issue Type	Action Taken	Count Removed
Extra whitespace	Trimmed	All

Algorithm Implementation Details

A. Apriori Algorithm

Apriori uses a level-wise breadth-first search to generate candidate itemsets. Apriori is slower for large datasets due to repeated scans of the transaction database.

Pseudocode:

1. Generate all 1-itemsets with support $\geq$ min_support
2. For $k = 2$ to max_size:
 - Generate candidate k-itemsets from frequent (k-1)-itemsets
 - Count support of candidates
 - Keep those $\geq$ min_support
3. Return all frequent itemsets

B. Eclat Algorithm

Eclat uses vertical format (TID-set) for improved efficiency. Eclat is generally faster than Apriori due to fewer full dataset scans.

Pseudocode:

1. Convert dataset into TID-lists
2. Recursively intersect TID-lists to produce larger itemsets
3. Keep itemsets with support $\geq$ min_support

Recommendations & Business Insights

After mining, the system lets users select a product and view associated items with confidence scores.

Example:

If a customer buys bread, they are likely to buy:

- Milk — 32%
- Butter — 24%

The system also generates business insights based on the strongest rule:

“Place Bread near Milk on shelves.

This association appears in 32% of matching transactions, indicating a strong cross-selling opportunity.”

Performance Analysis

Experiments were run on the cleaned dataset of 89 transactions. Performance was measured based on runtime, rule count, and structural efficiency.

- Apriori: Slower on larger datasets due to candidate explosion.
- Eclat: Faster due to vertical structure, but may use more memory.

Results:

Algorithm	Runtime (ms)	Rules Generated	Memory Usage
Apriori	40–90 ms	~15–40	Not measured
Eclat	15–40 ms	~15–40	Not measured

User Interface Design

The UI was built using Streamlit.

Components include:

- Transaction creation module
- CSV upload handler
- Preprocessing summary
- Algorithm selector
- Rule viewer
- Recommendation engine
- Business insights panel

Testing and Results

Test scenarios included checking data import, preprocessing correctness, rule output, and recommendation accuracy. Test transactions were used to produce predictable associations. The system consistently produced correct frequent itemsets and recommendations.

Conclusion

This project demonstrates how association rule mining can reveal product relationships in supermarket data. By integrating Apriori and Eclat into a user-friendly interface, the system provides valuable insights for cross-selling, product placement, and understanding consumer behavior. Streamlit allowed us to create

a quick UI development, while AI tools helped with debugging and explaining algorithmic concepts. Future improvements could include FP-Growth/CLOSET implementation, visualization graphs, and integration with larger datasets.